Derivatives of One Variable

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Where this fits: Pipeline map step 0 (Foundations). See the Course map.

0 Foundations	1 Frame	2 Get data	3 Understand	4 Clean	5 Features	6 Reduce
7 Split	8 Model	9 Evaluate	10 Tune	11 Deploy	12 Test	13 Monitor

- **Leads to:** Partial derivatives and gradients; Gradient descent; Maximum likelihood estimation (MLE); XGBoost; Backpropagation; Differential-drive kinematics; Kalman filter; Extended Kalman filter.

1. Overview

Key point: The derivative of f at x is the slope of the tangent line there: the **limit** (G-1087) of the slope of a secant line as its two points move together.

This Note follows Chapter 5 (Section 5.1) of *Mathematics for Machine Learning* (Deisenroth, Faisal and Ong, 2020).

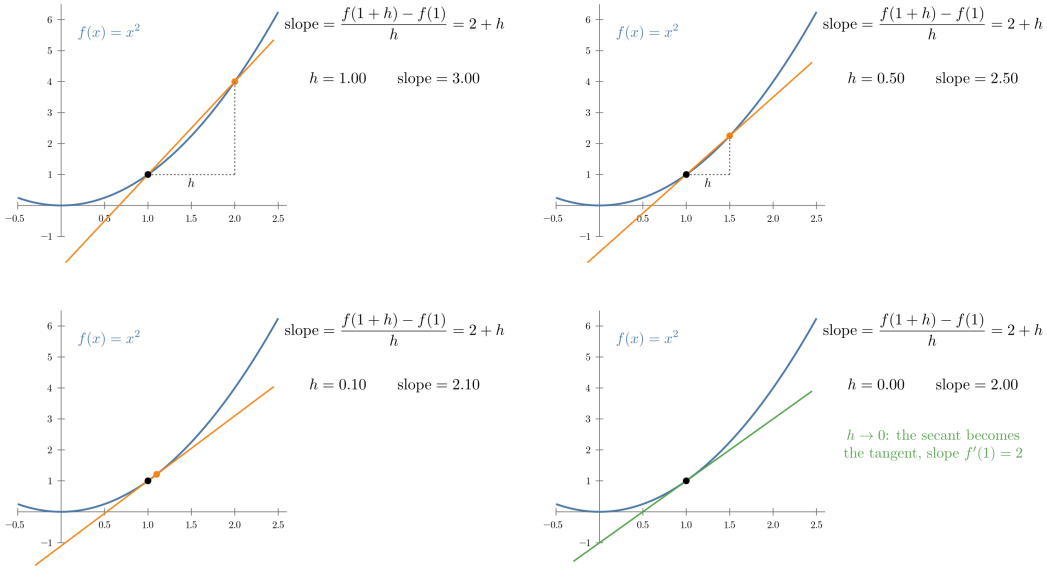


Figure 1: The secant line through $f(1)$ and $f(1+h)$ turns into the tangent line at $x = 1$ as h shrinks to 0

Figure 1 shows the whole idea of this Note. We draw a straight line through two points of a curve, a **secant line** (G-1758). As the second point slides towards the first, the secant line turns and settles on one line that only touches the curve: the **tangent line** (G-1945). Its slope is the **derivative** (G-595).

Earlier Notes already used derivatives as tools:

- the derivative as the slope of a **function** (G-816) at a point, set to zero in finding the minimum;

- stepping against the derivative in **gradient descent** (G-862);
- the **chain rule** (G-371) in the derivation of the sigmoid derivative;
- the **Taylor series** (G-1954) (a polynomial built from derivatives), in the Taylor series of XGBoost.

This Note explains where the derivative comes from (Sections 3 and 4), collects the rules for computing it (Section 5) and adds the **Taylor polynomial** (G-1953) (Section 6). The next Notes extend all of this to functions of many variables, which is what ML losses are.

2. Functions: one number in, one number out

> **Key point:** A function assigns exactly one output to every input. In this Note both are single numbers.

A **function** (G-816) f takes an input x and gives exactly one output $f(x)$. Our first example squares its input:

$$f(x) = x^2$$

$$f(3) = 3^2 = 9$$

$$f(-2) = (-2)^2 = 4$$

In this Note the input and the output are single real numbers. The short way to say so is

$$f : \mathbb{R} \to \mathbb{R}, \qquad x \mapsto f(x)$$

Each piece, read in words:

- $\mathbb{R}$ is the set of real numbers: every number on the number line, such as 3, -2 or 0.5;
- $f : \mathbb{R} \to \mathbb{R}$ says "a number goes in, a number comes out";
- $x \mapsto f(x)$ is read as: x maps to $f(x)$. It gives the rule; for our example, $x \mapsto x^2$ sends 3 to 9.

Figure 2 draws this map: each input on the top line sends exactly one arrow to the bottom line. Two inputs may land on the same output, as -2 and 2 both land on 4; what a function never does is send one input to two outputs.

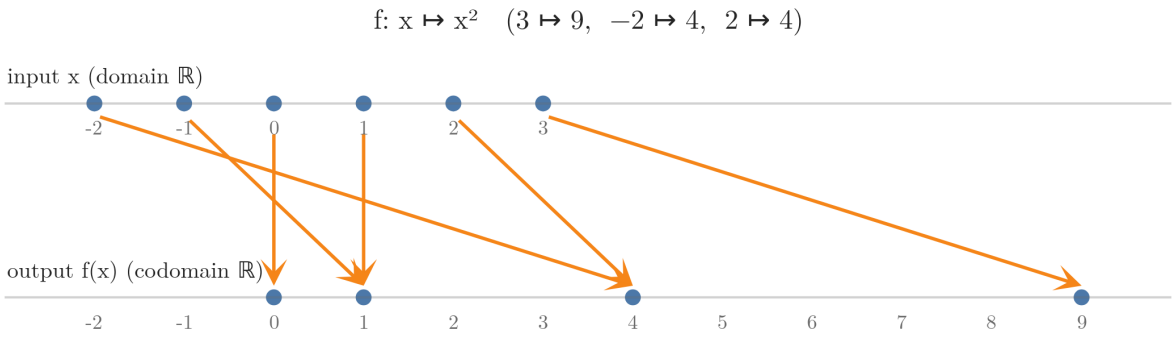


Figure 2: The function $f : x \mapsto x^2$ as a map from the input line to the output line, for the inputs -2 to 3. Every input has exactly one arrow; -2 and 2 share the output 4.

The set of allowed inputs is the **domain** (G-632); the set the outputs live in is the **codomain** (G-405). A loss is a number that scores how wrong a model is. ML losses take many numbers in (all the parameters) and give one number out. With n parameters this is written $f : \mathbb{R}^n \to \mathbb{R}$, where $\mathbb{R}^n$ means "lists of n numbers". For example, $\mathbb{R}^2$ holds pairs such as $(2, 3)$, and a loss of two parameters such as $f(x, y) = x^2 + y^2$ sends $(2, 3)$ to 13. Partial derivatives handle that case; this Note stays with one input.

3. The difference quotient: slope of a secant line

> **Key point:** Rise over run between two points of the curve, x and $x + h$, is the slope of the **secant line** (G-1758) through them: the average slope of f over that stretch.

Start with a car. It leaves a point A, speeds up, slows down and stops at a point B, 100 metres away, 10 seconds later. The top panel of Figure 3 plots the distance covered against time. The curve is shallow at the start (slow), steep in the middle (fast) and shallow again at the end.

How fast is the car going at exactly $t = 3$ seconds? A single instant gives nothing to measure: in one frozen moment the car covers no distance in no time. Speed needs **two** moments. A speedometer can take two moments very close together:

1. the car's position after t seconds is given by our curve $s(t)$ (in metres), with $s(3) = 21.600$ and $s(3.01) = 21.726$;
2. so in 0.01 s it moved
$$21.726 - 21.600 = 0.126 \text{ m}$$
3. and its speed is about
$$\frac{0.126}{0.01} = 12.6 \text{ metres per second}$$

The curve itself, with a helper u that is the time divided by 10:

$$u = t/10$$

$$s(t) = 100\,(3u^2 - 2u^3)$$

At $t = 3$, $u = 0.3$:

$$3u^2 = 3 \times 0.09 = 0.27$$

$$2u^3 = 2 \times 0.027 = 0.054$$

$$s(3) = 100 \times (0.27 - 0.054) = 21.6$$

Doing the same division at every t draws the whole speed curve, the bottom panel of Figure 3: low, high, low. "Change in distance divided by change in time" is a **rate of change** (G-1635), and the rest of this Note is about that one ratio.

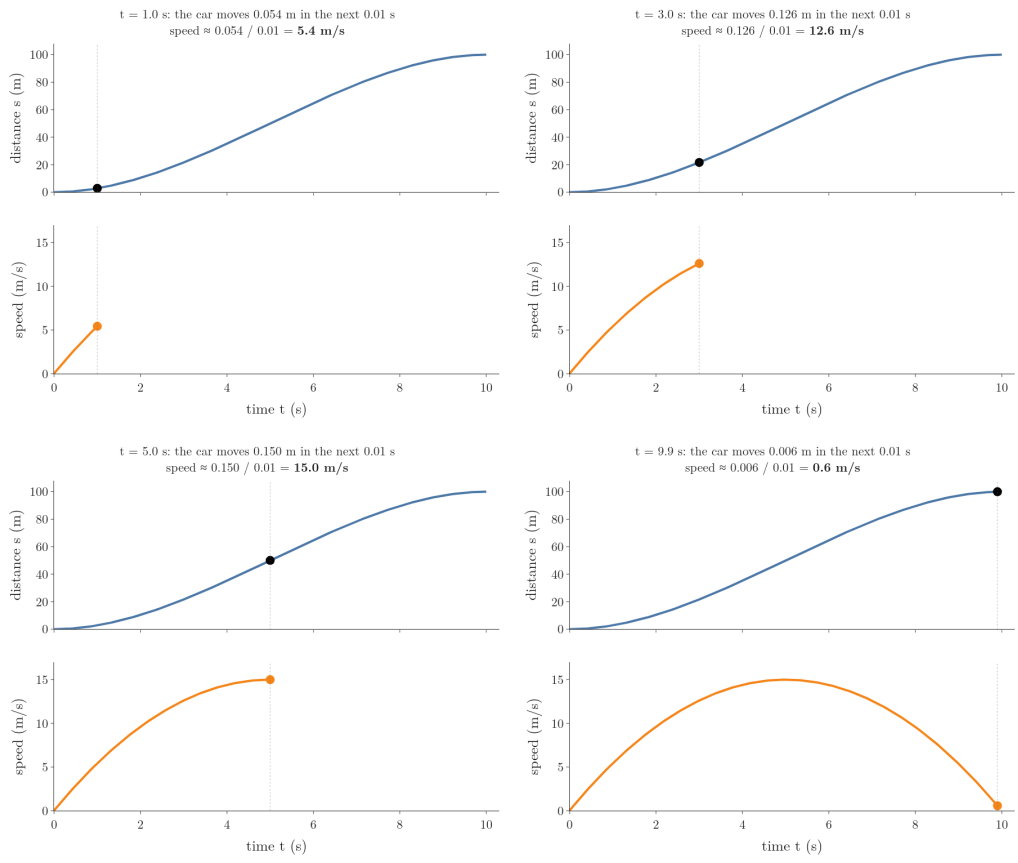


Figure 3: A car covers 100 m in 10 s. Top: distance against time. Bottom: the speed, built one point at a time as (distance moved in the next 0.01 s) / 0.01. Key frames: $t = 1$, 3, 5 and 9.9 s. Idea after 3Blue1Brown, "The paradox of the derivative"; our own curve

The same ratio works for any curve, with x in place of time and $f(x)$ in place of distance. Pick a point x on the curve and a second point a step h further. The secant line through the two points has a slope we can compute from the two heights.

1. **In words:** the change in output divided by the change in input.
2. **Formula:** write $y = f(x)$ for the output. The Greek capital Δ (delta) means "change in": $\Delta x = h$ is the change in the input and $\Delta y = f(x+h) - f(x)$ the change in the output. The **difference quotient** (G-604) is

$$\frac{\Delta y}{\Delta x} = \frac{f(x+h) - f(x)}{h}$$

3. **Example:** $f(x) = x^2$ at $x = 1$ with $h = 1$. The points are $(1, 1)$ and $(2, 4)$:

$$\frac{f(2) - f(1)}{1} = \frac{4 - 1}{1} = 3$$

The secant line has slope 3 (top-left frame of Figure 1). The secant slope is the slope a straight line would need to get from the first point to the second, so it is the **average slope** of f between x and $x + h$. A curve that bends in between can be steeper or flatter than this at any single point. Figure 4 shows both. The secant line rises 3 over a run of 1. The curve itself is flatter than that at $x = 1$ (tangent slope 2) and steeper at $x = 2$ (tangent slope 4); 3 is the average over the step.

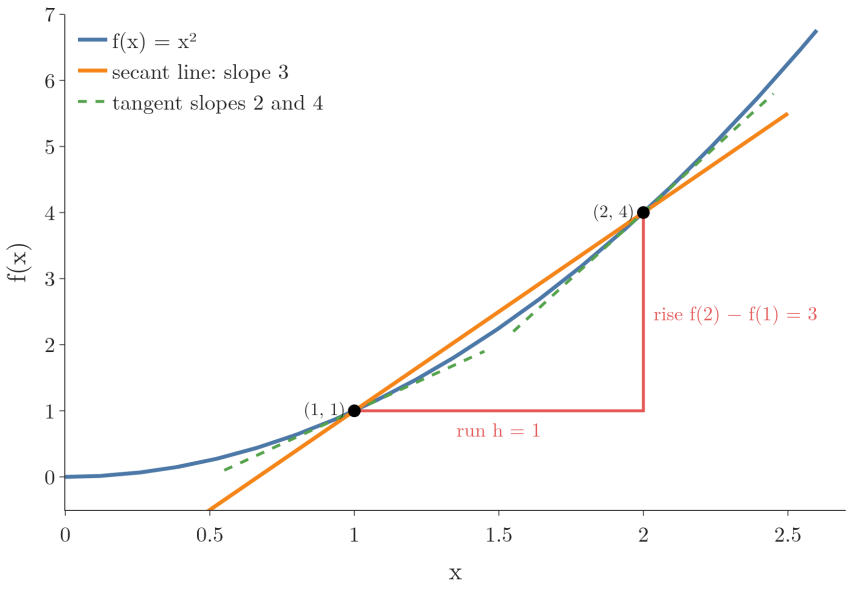


Figure 4: The difference quotient of f(x) = x² between x = 1 and x = 2 as rise over run: a rise of 3 over a run of 1, so the secant slope is 3. The dashed tangent lines at the two ends have slopes 2 and 4.

4. The derivative: shrinking the step to zero

> **Key point:** As $h \to 0$, the secant slope approaches one number, the slope of the tangent line; that **limit** (G-1087) is the derivative $f'(x)$.

4.1 From secant to tangent

> **Key point:** Make h smaller and smaller, and watch where the **difference quotient** (G-604) settles.

For $f(x) = x^2$ at $x = 1$, smaller steps give:

h	1	0.5	0.1	0.01	0.001
$\dfrac{(1+h)^2 - 1}{h}$	3	2.5	2.1	2.01	2.001

The slopes approach 2. Figure 1 shows the same thing as a picture: the **secant line** (G-1758) turns until, in the limit, it touches the curve at one point. That limiting line is the **tangent line**.

1. **In words:** the **derivative** (G-595) of f at x is the value the difference quotient approaches as the step h shrinks to zero.
2. **Formula:**
$$f'(x) = \frac{df}{dx} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$
$\lim_{h \to 0}$, the **limit** (G-1087), means "the value this approaches as h gets as close to 0 as we like". We never set $h = 0$ itself, which would give $0/0$.
3. **Example:** the algebra explains the table. For $f(x) = x^2$, one step per line:
$$(x+h)^2 = x^2 + 2xh + h^2$$
$$(x+h)^2 - x^2 = 2xh + h^2$$
$$\frac{2xh + h^2}{h} = 2x + h$$
$$2x + h \longrightarrow 2x \quad \text{as } h \to 0$$
At $x = 1$ the difference quotient is $2 + h$:
$$h = 1: \quad 2 + 1 = 3$$
$$h = 0.1: \quad 2 + 0.1 = 2.1$$
$$h \to 0: \quad f'(1) = 2$$

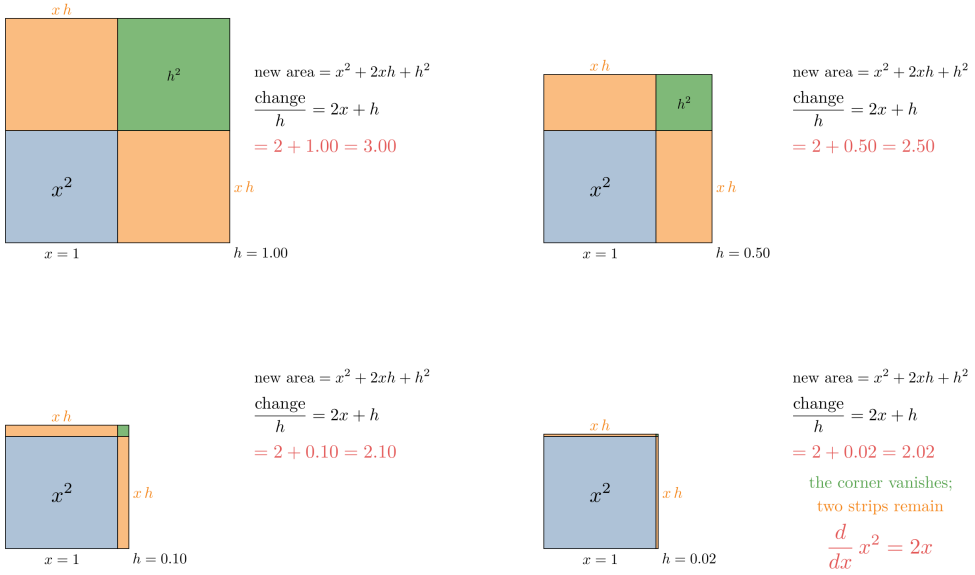


Figure 5: A square of side $x = 1$ grows by h. The new area is two orange strips of area xh and a green corner h^2, so the change divided by h is $2x + h$. Key frames: $h = 1$, 0.5, 0.1, and the limit

Figure 5 shows the same algebra as areas. Watch the green corner: it is h^2, so it shrinks much faster than the two strips, and only the strips, $2xh$, are left to divide by h. The picture is Sanderson's (3Blue1Brown, "**Power Rule** (G-1540) through geometry"), redrawn with our numbers.

A derivative is often called an "instantaneous **rate of change** (G-1635)". The car shows why that phrase needs care: nothing changes in a single instant. The derivative is the number the two-moment ratio settles on as the two moments close in. A safe reading is: the derivative is the best constant rate of change near the point. For the car at $t = 3$, the ratio settles on 12.6 m/s.

The two names $f'(x)$ and $\dfrac{df}{dx}$ mean the same thing. A **function** (G-816) whose derivative exists at a point is **differentiable** (G-606) there. **Gradient descent** (G-862) uses the derivative of the loss at every step, so it needs a loss that is differentiable.

4.2 The power rule from the definition

> **Key point:** Expanding $(x + h)^n$ shows that every term except one carries an h and vanishes, leaving $(x^n)' = nx^{n-1}$.

The same method works for $f(x) = x^3$, one step per line:

$$(x + h)^3 = x^3 + 3x^2h + 3xh^2 + h^3$$

$$(x+h)^3 - x^3 = 3x^2h + 3xh^2 + h^3$$

$$\frac{3x^2h + 3xh^2 + h^3}{h} = 3x^2 + 3xh + h^2$$

$$3x^2 + 3xh + h^2 \longrightarrow 3x^2 \quad \text{as } h \to 0$$

The x^3 cancels, one h divides out of every remaining term, and every term that still has an h goes to 0. Only $3x^2$ is left.

The picture is Figure 5 one dimension up. Read x^3 as the volume of a cube of side x. Growing the side by h adds three thin square slabs, one on each of three faces, each of volume x^2h. The remaining pieces are thin sticks and one tiny corner, all with h^2 or h^3 in them. Divide by h and let h shrink: only the three slabs survive, $3x^2$.

1. **In words:** the **power rule** (G-1540): bring the power down in front and lower the power by one.
2. **Formula:**
$$\frac{d}{dx}x^n = n\,x^{n-1}$$
3. **Example:** for $f(x) = x^3$:
$$f'(x) = 3x^2$$
$$f'(2) = 3 \times 2^2 = 3 \times 4 = 12$$

> **Extra:** For a general n, the binomial theorem writes
>
> $$(x+h)^n = x^n + nx^{n-1}h$$
> $$+(\text{terms with } h^2, h^3, \dots)$$
>
> Subtracting x^n and dividing by h leaves nx^{n-1} plus terms that all still contain h, so the **limit** (G-1087) is nx^{n-1}, exactly as for $n = 3$.

4.3 What the sign of the derivative tells us

> **Key point:** A positive **derivative** (G-595) means f rises as x grows; a negative one means it falls. So the derivative points uphill.

The derivative is the **rate of change** (G-1635) of the output per unit of input, near x. Its sign says which way is uphill:

- $f'(x) > 0$: increasing x increases f;
- $f'(x) < 0$: increasing x decreases f;
- $f'(x) = 0$: the tangent is flat, as at the bottom of a valley.

The sign is all **gradient descent** (G-862) needs: to go downhill, move x against the sign of the derivative. For $f(x) = x^2$ at $x = 1$, $f'(1) = 2 > 0$, so we move left, towards the minimum at 0.

Python: A computer can estimate a derivative with a small h, straight from the definition. Such an estimate is a **numerical derivative** (G-1368; or finite difference).

```
import numpy as np

def f(x):
    return x ** 3

h = 0.1
(f(1 + h) - f(1)) / h              # 3.31
(f(1 + h) - f(1 - h)) / (2 * h)    # 3.01
```

The exact answer is $f'(1) = 3$.

Extra: The second line is the **central difference** (G-363): it uses one point on each side of x. Its error shrinks like h^2 instead of h, so with $h = 0.1$ it is off by 0.01 instead of 0.31. Making h tiny (say 10^{-12}) does not help: the two heights become almost equal and rounding errors in the computer take over. Balancing the two errors puts the best h for the central difference near the cube root of the machine precision (the smallest relative rounding step of the computer):

$$(2.2 \times 10^{-16})^{1/3} \approx 6 \times 10^{-6}$$

So values near $h = 10^{-5}$ are a common choice (Nocedal and Wright §8.1).

5. Rules for computing derivatives

Key point: We rarely take **limits** (G-1087) by hand: a short table of basic derivatives plus four rules (sum, product, quotient, chain) gives the derivative of any formula built from them.

5.1 Basic derivatives

Key point: These few **derivatives** (G-595), combined by the rules below, cover the functions ML uses.

$f(x)$	$f'(x)$	Where ML meets it
c (a constant)	0	terms that do not depend on the parameter
x^n	nx^{n-1}	squared error
e^x	e^x	sigmoid, softmax (functions that turn scores into probabilities)

$f(x)$	$f'(x)$	Where ML meets it
$\ln x$	$1/x$	log loss
$\sin x$	$\cos x$	
$\cos x$	$-\sin x$	

Each entry can be proved from the **limit** (G-1087) definition, as we did for x^n. Two of them have a short picture proof.

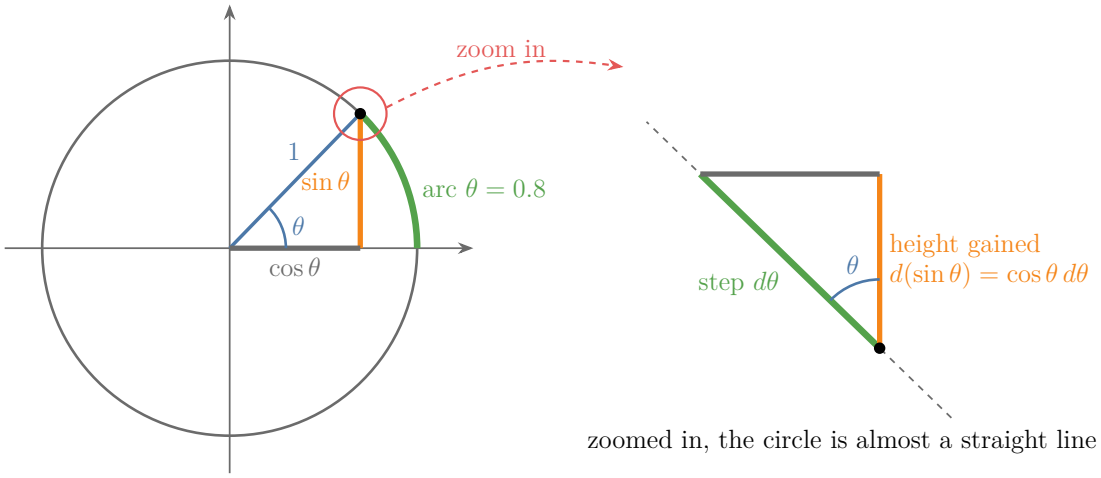
Why the derivative of sin **is** cos**.** Walk a distance θ around a circle of radius 1, starting from its rightmost point. The height of the point reached is $\sin\theta$ (left of Figure 6). Now take one more tiny step $d\theta$ along the circle.

1. Zoomed in, the circle is almost a straight line, so the step is the long side of a tiny right-angled triangle (right of Figure 6).
2. The tiny triangle has the same angles as the big one, so its upright side is $\cos\theta$ times its long side.
3. The upright side is the height gained: $d(\sin\theta) = \cos\theta\, d\theta$. Dividing by $d\theta$ gives $\cos\theta$.

At $\theta = 0.8$, with a step $d\theta = 0.01$:

$$\text{slope} = \cos 0.8 = 0.697$$

$$\text{height gained} \approx 0.697 \times 0.01 = 0.007$$



$$\frac{d(\sin\theta)}{d\theta} = \frac{\text{side next to } \theta}{\text{hypotenuse}} = \cos\theta$$

Figure 6: Left: a walk of length $\theta = 0.8$ around the unit circle; $\sin\theta$ is the height reached. Right: one more tiny step $d\theta$, zoomed in. The tiny triangle has the same angles as the big one, so the height gained is $\cos\theta\, d\theta$. Idea after 3Blue1Brown, "Derivative formulas through geometry"

Why the derivative of $1/x$ is $-1/x^2$. Picture a rectangular puddle whose area is always 1. If its width is x, its height must be $1/x$: width 2 forces height $\frac{1}{2}$, width 3 forces height $\frac{1}{3}$. Widen the puddle by a tiny dx:

1. the new strip on the side has height $1/x$ and width dx, so it adds an area of about
$$\frac{1}{x}\,dx$$
2. the area must stay 1, so the top must drop. Call the change in height $d(1/x)$; the lost layer has width x, so its area is $x \cdot d(1/x)$, and it must cancel the strip:
$$x \cdot d\left(\frac{1}{x}\right) = -\frac{1}{x}\,dx$$
3. divide both sides by $x\,dx$:
$$\frac{d(1/x)}{dx} = -\frac{1}{x^2}$$

At $x = 2$:

$$-\frac{1}{2^2} = -\frac{1}{4} = -0.25$$

The **power rule** (G-1540) agrees: $1/x = x^{-1}$, so the derivative is $-1 \cdot x^{-2}$.

5.2 Sum, product and quotient rules

> **Key point: Derivatives** (G-595) of a sum add up; a product and a quotient have their own fixed patterns.

Each rule combines two **functions** (G-816), f and g, whose derivatives f' and g' we already know. Each rule below names its own f and g with numbers first.

Sum rule: the derivative of a sum is the sum of the derivatives.

Take $f(x) = x^2$ and $g(x) = x^3$, at $x = 1$:

$$f'(x) = 2x$$

$$f'(1) = 2$$

$$g'(x) = 3x^2$$

$$g'(1) = 3$$

$$(x^2 + x^3)' \text{ at } x = 1: \quad 2 + 3 = 5$$

In general:

$$\Big(f(x) + g(x)\Big)' = f'(x) + g'(x)$$

Product rule (G-1577): differentiate one factor at a time, keep the other, and add.

Take $f(x) = x^2$ and $g(x) = 3x + 1$, and their product $x^2(3x + 1)$, at $x = 1$. The four pieces:

$$f(1) = 1^2 = 1$$

$$f'(x) = 2x$$

$$f'(1) = 2$$

$$g(1) = 3 + 1 = 4$$

$$g'(x) = 3$$

$$g'(1) = 3$$

The first product (change f, keep g):

$$f'(1)\, g(1) = 2 \times 4 = 8$$

The second product (keep f, change g):

$$f(1)\, g'(1) = 1 \times 3 = 3$$

Add them:

$$8 + 3 = 11$$

Check by multiplying out first:

$$x^2(3x + 1) = 3x^3 + x^2$$

$$(3x^3 + x^2)' = 9x^2 + 2x$$

$$9 \times 1 + 2 \times 1 = 11$$

The same. In general:

$$\Big(f(x)\, g(x)\Big)' = f'(x)\, g(x) + f(x)\, g'(x)$$

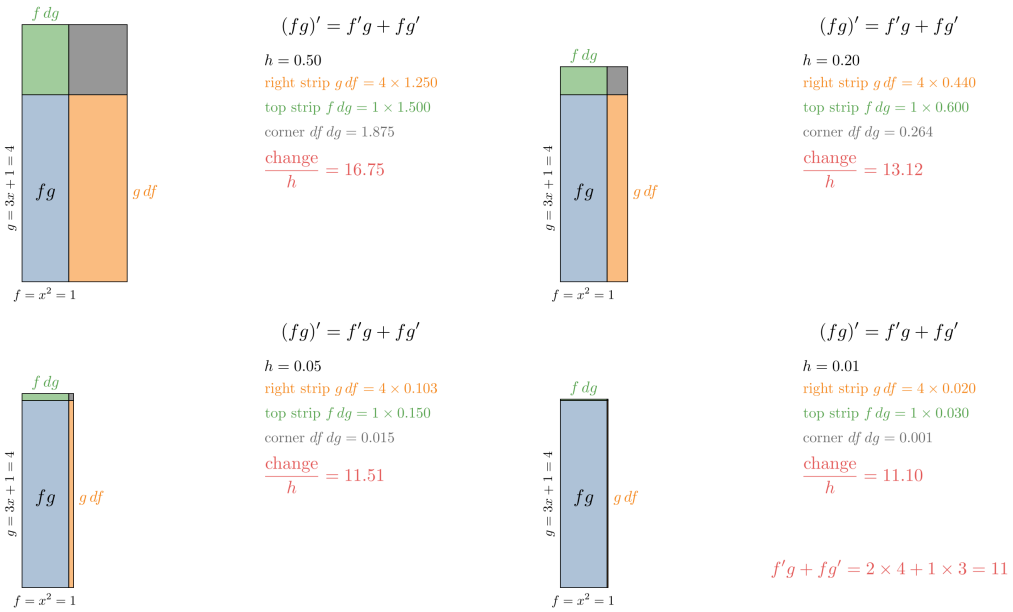


Figure 7: The product fg as the area of a rectangle with sides $f = x^2$ and $g = 3x + 1$ (1 by 4 at $x = 1$). Nudging x by h adds a right strip $g\,df$, a top strip $f\,dg$ and a tiny corner; the change divided by h settles at 11

Figure 7 draws the product rule as a growing rectangle, again after Sanderson (3Blue1Brown, "Visualizing the **chain rule** (G-371) and product rule"). The sides are $f = x^2$ and $g = 3x+1$, so the area is the product. Watch the corner $df\,dg$ vanish: what remains is one strip per factor, the 8 and the 3 above.

Quotient rule (G-1609): for a fraction.

Take $f(x) = x$ on top and $g(x)$ underneath, at $x = 2$:

$$g(x) = x^2 + 1$$

$$f(2) = 2$$

$$f'(x) = 1$$

$$f'(2) = 1$$

$$g(2) = 4 + 1 = 5$$

$$g'(x) = 2x$$

$$g'(2) = 4$$

$$f'(2)\,g(2) = 1 \times 5 = 5$$

$$f(2)\,g'(2) = 2 \times 4 = 8$$

$$g(2)^2 = 5^2 = 25$$

$$\left(\frac{x}{x^2+1}\right)' \text{ at } x = 2:$$

$$\frac{5-8}{25} = \frac{-3}{25}$$

$$= -0.12$$

In general:

$$\left(\frac{f(x)}{g(x)}\right)' = \frac{f'(x)\,g(x) - f(x)\,g'(x)}{g(x)^2}$$

5.3 The chain rule

> **Key point:** For a function of a function, multiply the outer **derivative** (G-595) (evaluated at the inside) by the inner derivative.

Start with two straight lines (Figure 8). Suppose weight predicts height, and height predicts shoe size:

- Height is 2 times weight, so one more unit of weight gives 2 more units of height:

$$\text{height} = 2 \times \text{weight}$$

- Shoe size is one quarter of height, so one more unit of height gives a quarter of a unit more shoe size:

$$\text{shoe size} = \tfrac{1}{4} \times \text{height}$$

How much does shoe size change per unit of weight? Follow the change through the middle quantity, one link per line:

$$\text{weight: } +1$$

$$\text{height:} \quad +2 \times 1 = +2$$

$$\text{shoe size:} \quad +\tfrac{1}{4} \times 2 = +\tfrac{1}{2}$$

Height links weight to shoe size, so the two slopes multiply:

$$\frac{d\,\text{shoe}}{d\,\text{weight}}$$

$$= \frac{d\,\text{shoe}}{d\,\text{height}} \times \frac{d\,\text{height}}{d\,\text{weight}}$$

$$= \frac{1}{4} \times 2 = \frac{1}{2}$$

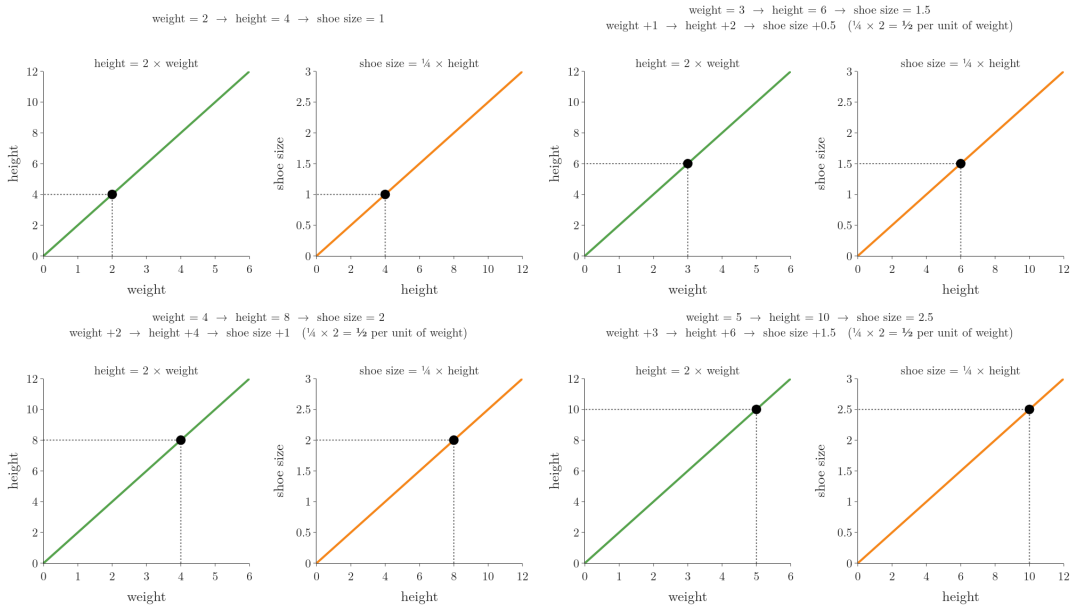


Figure 8: Two lines chained together: height = 2 × weight (left) and shoe size = ¼ × height (right). As the weight moves from 2 to 5, the height moves twice as far and the shoe size half as far as the weight: ¼ × 2 = ½. Numbers after StatQuest, "The Chain Rule, Clearly Explained!!!"

In Figure 8, watch the two dots: each unit of weight moves the left dot up by 2 and the right dot up by $\frac{1}{2}$. Multiplying the rates along the chain is the **chain rule** (G-371). The sigmoid derivative used it: to differentiate a function of a **function** (G-816), multiply the outer derivative by the inner derivative. We write it once more with the notation of **composition** (G-431): $g \circ f$ means "first f, then g", so $(g \circ f)(x) = g(f(x))$.

1. **In words:** differentiate the outer function, leaving the inside untouched, then multiply by the derivative of the inside.
2. **Formula:**
$$(g \circ f)'(x) = g'\Big(f(x)\Big)\, f'(x)$$
3. **Example:** take
$$h(x) = (x^2 + 1)^3$$
At $x = 1$:
$$h(1) = 2^3 = 8$$
Split it into an inside and an outside:
$$\text{inside: } f(x) = x^2 + 1$$
$$f'(x) = 2x$$
$$\text{outside: } g(u) = u^3$$
$$g'(u) = 3u^2$$
At $x = 1$, one step per line:
$$f(1) = 1 + 1 = 2$$
$$f'(1) = 2 \times 1 = 2$$
$$g'(f(1)) = g'(2) = 3 \times 2^2 = 12$$
$$h'(1) = 12 \times 2 = 24$$
In general:
$$h'(x) = 3(x^2 + 1)^2 \cdot 2x$$

The rates multiply here exactly as the two slopes did in Figure 8; the only change is that the rates of a curve depend on the point. Near $x = 1$, read the lines above as rates:

- the inside changes 2 times as fast as x;
- the outside changes 12 times as fast as the inside;
- so h changes 12 times 2, that is 24 times, as fast as x.

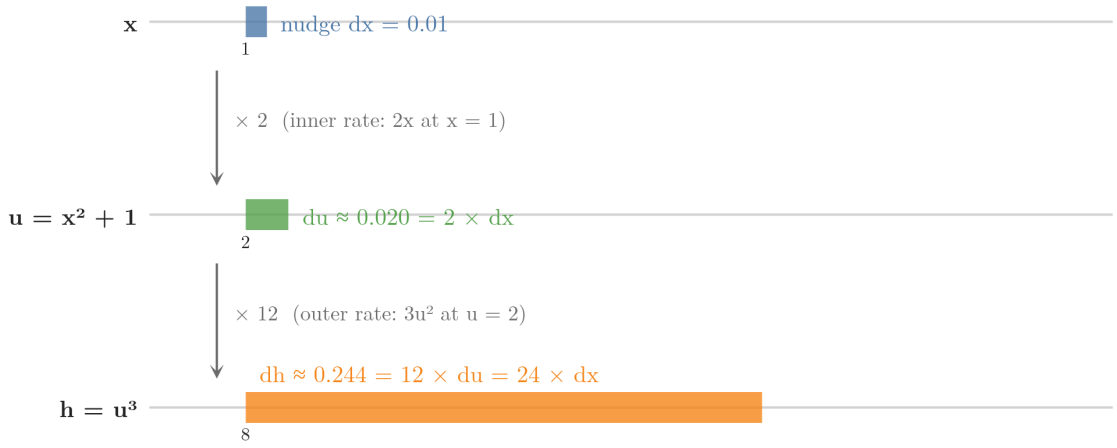


Figure 9: The chain rule for $h(x) = (x^2 + 1)^3$ at $x = 1$ as three number lines. A nudge $dx = 0.01$ in x becomes a nudge of about 0.02 in the inside $u = x^2 + 1$ (rate 2), and about 0.24 in $h = u^3$ (rate 12 on top): 24 times dx. After Sanderson (3Blue1Brown), "Visualizing the chain rule and product rule".

Figure 9 draws these rates as nudges on three number lines, one per stage of the chain. Each stage stretches the nudge it receives by its own rate, so the stretches multiply.

Most ML models are long chains of functions: a weighted sum, then a sigmoid, then a log loss. The chain rule is what turns their derivatives into a product of small, easy pieces. Finding the minimum already used it on $(y_i - mx_i - b)^2$, and the next Notes extend it to many variables.

The smallest case of that use fits in three lines. One person has weight 2 and height 3. We predict height as $b + 1\times$ weight and may only move the intercept b. The error on this person is the **residual** (G-705):

$$r = 3 - (b + 1 \times 2) = 1 - b$$

For example, $b = 0$ gives $r = 1$. The loss is r^2.

1. Outer rate: $\dfrac{d(r^2)}{dr} = 2r$.
2. Inner rate: $\dfrac{dr}{db} = -1$ (raising the intercept by 1 lowers the residual by 1).
3. Chain rule: multiply the two rates.

$$\frac{d(r^2)}{db} = 2r \times (-1)$$

$$\frac{d(r^2)}{db} = -2(1 - b)$$

The loss is lowest where this derivative is 0, at $b = 1$: the line then passes through the person's point.

> **Python:** Checking a rule numerically is a good habit.
> ```
> h_fun = lambda x: (x ** 2 + 1) ** 3
> step = 1e-5
> (h_fun(1 + step) - h_fun(1 - step)) / (2 * step)    # 24.0000...
> ```

6. Taylor polynomials

> **Key point:** Near a point x_0, a function is approximated by a polynomial built from its **derivatives** (G-595) at x_0; the degree-1 polynomial is the **tangent line** (G-1945).

The Taylor series in XGBoost introduced the **Taylor series** (G-1954): near a point, a smooth **function** (G-816) is approximated by a **polynomial** (a sum of powers of x such as $1 - \frac{1}{2}x^2$) built from its value and derivatives there. It worked through e^x, adding terms up to $x^3/6$. Here we name the pieces and add what is new.

6.1 Building one by hand: match the value, the slope and the bend

> **Key point:** Choose a polynomial's coefficients one at a time so that it has the same value, the same slope and the same bend as the **function** (G-816) at one point.

Take $\cos x$ near $x = 0$. We want a simple polynomial $c_0 + c_1 x + c_2 x^2$ that stays as close to $\cos x$ as possible near 0. Figure 10 builds it in steps.

1. **Match the value.** $\cos 0 = 1$, and the polynomial at 0 is c_0. So $c_0 = 1$.
2. **Match the slope.** The slope of $\cos x$ at 0 is $-\sin 0 = 0$: the curve is flat at its top. The slope of the polynomial at 0 is c_1. So $c_1 = 0$.
3. **Match the bend.** The cosine curves downwards at 0. The **second derivative** (G-2249), the **derivative** (G-595) of the derivative, measures this bend: for $\cos x$ it is $-\cos 0 = -1$. For the polynomial it is $2c_2$. So $2c_2 = -1$ and $c_2 = -\frac{1}{2}$.

The result is

$$\cos x \approx 1 - \tfrac{1}{2}x^2$$

Check at $x = 0.1$:

$$\tfrac{1}{2} \times 0.1^2 = \tfrac{1}{2} \times 0.01 = 0.005$$

$$1 - 0.005 = 0.995$$

The true $\cos 0.1$ is 0.99500 to five places.

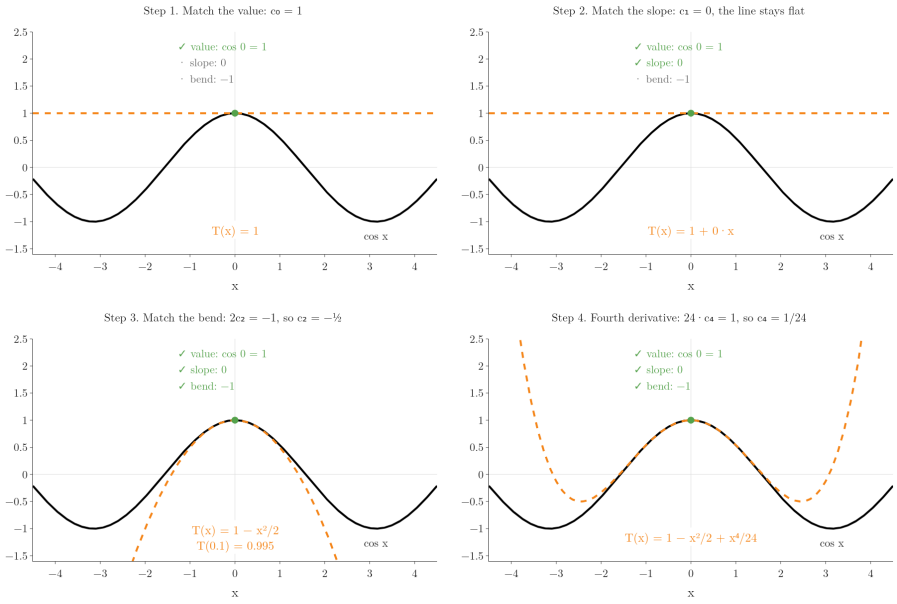


Figure 10: Building the Taylor polynomial of $\cos x$ at 0. Step 1 matches the value, step 2 the slope, step 3 the bend, giving $1 - x^2/2$; step 4 adds $x^4/24$, which follows the curve for longer. Idea after 3Blue1Brown, "Taylor series"

One more term makes the fit last longer (step 4 of Figure 10). The fourth derivative of $\cos x$ at 0 is 1. Differentiate $c_4 x^4$ four times; each derivative brings one power down:

$$\text{first: } 4c_4x^3$$

$$\text{second: } 4 \cdot 3\, c_4x^2 = 12c_4x^2$$

$$\text{third: } 12 \cdot 2\, c_4x = 24c_4x$$

$$\text{fourth: } 24c_4$$

Matching $24c_4 = 1$ gives $c_4 = \frac{1}{24}$. (The x^3 coefficient is 0, because the third derivative of $\cos x$ at 0 is $\sin 0 = 0$.)

Two patterns stand out:

- each derivative at 0 is controlled by exactly one coefficient, so adding a new term never spoils the earlier matches;
- the k-th coefficient is the k-th derivative divided by $1 \cdot 2 \cdots k$, to cancel the powers brought down.

The general formula of Section 6.2 is these two patterns written once for every k.

6.2 The Taylor polynomial of degree n

> **Key point:** Keep the terms up to $(x - x_0)^n$ of the **Taylor series** (G-1954): that finite sum is the Taylor polynomial T_n.

We write $f^{(k)}$ for the k-th **derivative** (G-595): $f^{(0)} = f$, $f^{(1)} = f'$, $f^{(2)} = f''$, and so on. The number $k!$ is called k factorial, with $0! = 1$:

$$k! = 1 \cdot 2 \cdots k$$

For example:

$$3! = 1 \cdot 2 \cdot 3 = 6$$
$$5! = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 = 120$$

The symbol $\sum_{k=0}^{n}$ (capital sigma) means "add up the terms for $k = 0, 1, 2, \ldots, n$".

1. **In words:** for each k from 0 to n, take the k-th derivative at x_0, divide by $k!$ and multiply by $(x - x_0)^k$; add these up.
2. **Formula:** the **Taylor polynomial** (G-1953) of degree n at x_0 is

   $$T_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k$$

   Letting n run to infinity gives the Taylor series T_∞. With $x_0 = 0$ it is called the **Maclaurin series** (G-1142).
3. **Example:** $f(x) = \sin x$ at $x_0 = 0$. The derivatives cycle $\sin, \cos, -\sin, -\cos$, which at 0 give $0, 1, 0, -1$. So only odd powers survive:

   $$T_5(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!}$$
   $$T_5(x) = x - \frac{x^3}{6} + \frac{x^5}{120}$$

   At $x = 0.5$, the three terms are:

   $$x = 0.5$$
   $$\frac{x^3}{6} = \frac{0.125}{6} = 0.02083$$
   $$\frac{x^5}{120} = \frac{0.03125}{120} = 0.00026$$

   Adding one term at a time:

   $$T_1 = 0.5$$
   $$T_3 = 0.5 - 0.02083 = 0.47917$$
   $$T_5 = 0.47917 + 0.00026 = 0.47943$$

   The true $\sin 0.5 = 0.47943$.

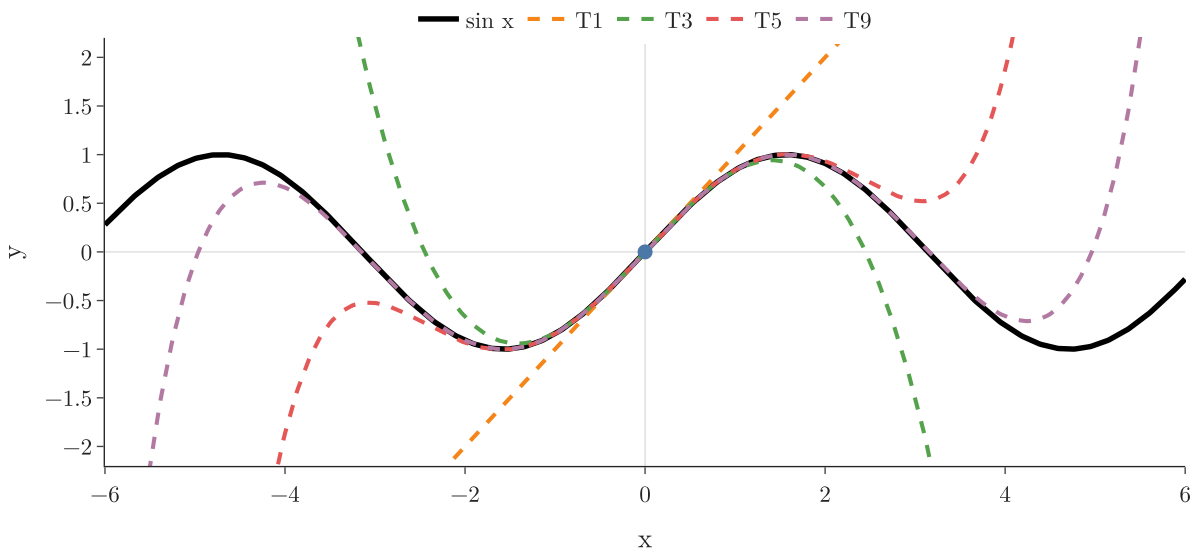


Figure 11: Taylor polynomials of $\sin x$ around $x_0 = 0$ (blue dot): each higher degree follows the curve over a wider range

Figure 11 shows the pattern. Near x_0 every polynomial is close to $\sin x$. Further away, the low-degree ones drift off, while T_9 stays on the curve for a whole wave. A Taylor polynomial is a **local** approximation: good near x_0, getting worse with distance.

6.3 Degree 1: the tangent line

> **Key point:** $T_1(x) = f(x_0) + f'(x_0)(x - x_0)$ is the **tangent line** (G-1945) at x_0; using it in place of f is called linearisation.

The first two terms are

$$T_1(x) = f(x_0) + f'(x_0)\,(x - x_0)$$

a straight line through $(x_0, f(x_0))$ with slope $f'(x_0)$: the tangent line of Figure 1. Replacing a **function** (G-816) by its tangent line near a point is called **linearisation** (G-1099).

1. **In words:** start at the known height and follow the slope for the distance moved.
2. **Formula:** for x near x_0,

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0)$$

3. **Example:** $f(x) = \sqrt{x}$ at $x_0 = 4$:

$$f(4) = \sqrt{4} = 2$$

$$f'(x) = \frac{1}{2\sqrt{x}}$$

$$f'(4) = \frac{1}{2 \times 2} = 0.25$$

$$\sqrt{4.1} \approx 2 + 0.25 \times 0.1 = 2.025$$

$$\text{true: } \sqrt{4.1} = 2.0248$$

$$\sqrt{5} \approx 2 + 0.25 \times 1 = 2.25$$

$$\text{true: } \sqrt{5} = 2.2361$$

A step of 0.1 is almost exact; a step of 1 is already visibly off.

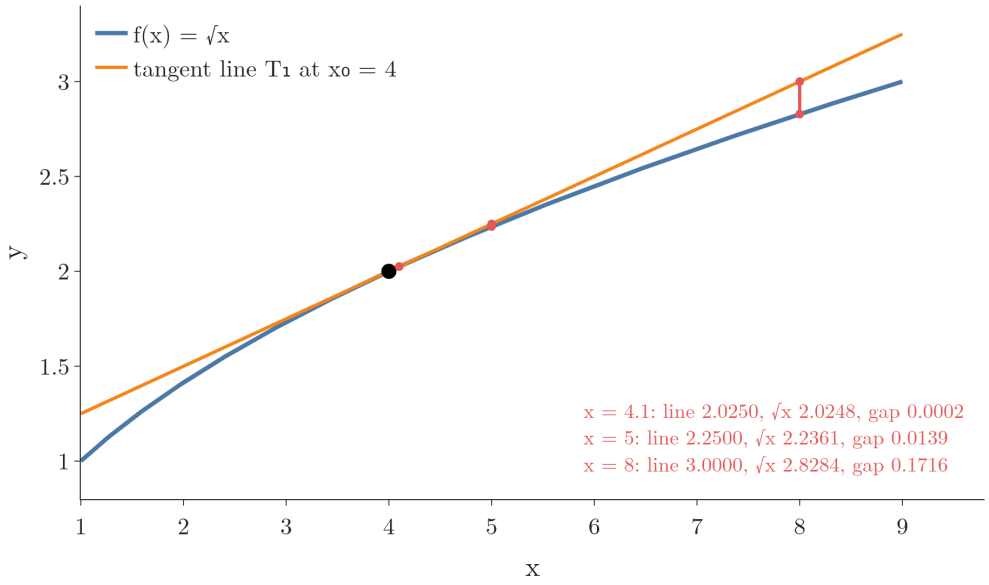


Figure 12: Linearisation of $\sqrt{x}$ at $x_0 = 4$: the tangent line (orange) against the curve (blue). The red gaps grow with the distance from x_0: 0.0002 at x = 4.1, 0.0139 at x = 5, 0.1716 at x = 8.

Figure 12 shows why. The tangent line hugs the curve near $x_0 = 4$, and the curve bends away from it more and more as we move further out.

Linearisation is the picture behind **gradient descent** (G-862). Each step trusts the tangent line, which is only reliable near the current point. A small learning rate (the size of each step, see how far to move) keeps the step inside the region where the tangent line is a good guide; with a step that is too large, gradient descent can overshoot and fail to converge (MML §7.1.1).

6.4 A polynomial is its own Taylor polynomial

> **Key point:** For a polynomial of degree k, all **derivatives** (G-595) beyond the k-th are zero, so T_n with $n \geq k$ reproduces it exactly.

For a **function** (G-816) that is not a polynomial, such as $\sin x$, a **Taylor polynomial** (G-1953) is only an approximation. For a polynomial it can be exact.

1. **In words:** a polynomial of degree 3 has a zero fourth derivative, so the **Taylor series** (G-1954) stops after the cube term and nothing is lost.
2. **Formula:** if f is a polynomial of degree k, then $T_n = f$ for every $n \geq k$.
3. **Example:** $f(x) = x^3$ at $x_0 = 2$. The derivatives at 2, one per line:

$$f(2) = 2^3 = 8$$

$$f'(x) = 3x^2$$
$$f'(2) = 12$$
$$f''(x) = 6x$$
$$f''(2) = 12$$
$$f'''(x) = 6$$
$$f'''(2) = 6$$

All higher derivatives are 0. So

$$T_3(x) = 8 + 12(x-2)$$
$$+\frac{12}{2!}(x-2)^2 + \frac{6}{3!}(x-2)^3$$
$$T_3(x) = 8 + 12(x-2)$$
$$+6(x-2)^2 + (x-2)^3$$

Multiply out each bracket:

$$12(x-2) = 12x - 24$$
$$6(x-2)^2 = 6x^2 - 24x + 24$$
$$(x-2)^3 = x^3 - 6x^2 + 12x - 8$$

Collect the terms by power of x:

$$\text{constant: } 8 - 24 + 24 - 8 = 0$$

$$x: \quad 12 - 24 + 12 = 0$$
$$x^2: \quad 6 - 6 = 0$$
$$x^3: \quad 1$$

So $T_3(x) = x^3$ exactly.

Exactness for polynomials explains a result of the second-order approximation of the objective: its second-order approximation of the squared error was exact, because the squared error is already a polynomial of degree 2.

> **Extra:** A function equal to its Taylor series everywhere near x_0 is called **analytic** (G-197); e^x, $\sin x$ and $\cos x$ are, at every point. A Taylor series is one example of a **power series** (G-1541), $\sum a_k(x-c)^k$, a polynomial with infinitely many terms. Functions such

as `np.sin` and `np.exp` are computed with polynomial approximations too (Muller 2016).

Extra: Degree 2 is where the curvature enters: T_2 is the parabola that matches the value, the slope and the **second derivative** (G-2249) at x_0. Jumping to the lowest point of that parabola is the **Newton step** of leaf values in log-odds. The quadratic approximation does the same with many variables.

7. Summary

Idea	Formula	Example ($f(x) = x^2$ at $x = 1$ unless stated)	Why it matters
Difference quotient	$(f(x+h) - f(x))/h$	$h = 0.1$: 2.1	lets a computer estimate a derivative numerically
Derivative	limit of the difference quotient as $h \to 0$	$f'(1) = 2$	the slope at one point; its sign says which way is uphill
Power rule	The rule for differentiating a power of x: bring the power down in front and lower the power by one, $(x^n)' = nx^{n-1}$.	$(x^3)' = 3x^2$	the derivative of a power with no limit to take
Chain rule	$(g \circ f)' = g'(f(x))\, f'(x)$	$((x^2+1)^3)'$ at $1 = 24$	differentiates a function of a function, such as a loss of a residual
Taylor polynomial	sum of $f^{(k)}(x_0)(x - x_0)^k/k!$ up to $k = n$	$\sin 0.5 \approx T_5 = 0.47943$	swaps a hard function for a polynomial near a point
Linearisation	$f(x_0) + f'(x_0)(x - x_0)$	$\sqrt{4.1} \approx 2.025$	the picture behind each gradient-descent step; reliable only near x_0

- The difference quotient is the slope of a secant line; its **limit** (G-1087) as $h \to 0$ is the slope of the tangent line, the derivative, so the derivative is the **rate of change** (G-1635) of f at one point.
- The sign of the **derivative** (G-595) says which way is uphill; gradient descent moves the other way, so the sign alone tells it which way to step.
- Sum, product, quotient and **chain rules** (G-371) build the derivative of any formula from a short table, so we rarely take limits by hand.
- A Taylor polynomial approximates a function near a point; degree 1 is the **tangent line** (G-1945), and a polynomial is reproduced exactly, because its derivatives beyond its degree are zero.
- So the derivative is the slope of the tangent line, found as the limit of secant slopes, and it tells an optimiser which way f goes up.

8. Sources

Built from

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- 3Blue1Brown, "Visualizing the chain rule and product rule | Chapter 4, Essence of calculus", YouTube, https://www.youtube.com/watch?v=YG15m2VwSjA
- 3Blue1Brown, "Taylor series | Chapter 11, Essence of calculus", YouTube, https://www.youtube.com/watch?v=3d6DsjIBzJ4
- StatQuest with Josh Starmer, "The Chain Rule, Clearly Explained!!!", YouTube, https://www.youtube.com/watch?v=wl1myxrtQHQ

Other references

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9. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Limit (G-1087)	The value an expression approaches as a quantity (such as h) gets arbitrarily close to a target (such as 0).
Secant line (G-1758)	A straight line through two points of a curve; as the second point slides towards the first, it turns into the tangent line, whose slope is the derivative.
Tangent line (G-1945)	The line that touches a curve at one point with the curve's slope there; the limit of secant lines.
Derivative (G-595)	The slope of a function at a point: how fast the output changes as the input changes there. It is zero at the lowest point of a smooth curve, which is how training finds a minimum.

Term	Meaning
Function (G-816)	A rule that assigns exactly one output to every input, written $f : \mathbb{R} \to \mathbb{R}$, $x \mapsto f(x)$.
Taylor polynomial (G-1953)	A polynomial built from a function's value and its first n derivatives at one point x_0, used to approximate the function near x_0; it is the Taylor series cut after the $(x - x_0)^n$ term.
Domain (G-632)	The set of allowed inputs of a function.
Codomain (G-405)	The set in which a function's outputs lie.
Rate of change (G-1635)	How much one quantity changes per unit change of another; the meaning of a derivative.
Difference quotient (G-604)	The average slope of a function over a step h: the change in output divided by the change in input, $(f(x + h) - f(x))/h$, the slope of the secant line. Shrinking h towards 0 gives the derivative.
Power rule (G-1540)	The rule for differentiating a power of x: bring the power down in front and lower the power by one, $(x^n)' = nx^{n-1}$.
Differentiable (G-606)	Having a derivative (one clear slope) at a point, or at every point; the absolute value $\|m\|$ is not, at its corner $m = 0$. Gradient descent needs a differentiable loss.
Numerical derivative (G-1368)	An estimate of a derivative straight from its definition: the change in the function over a small step h, divided by h (a finite difference); a computer can get it from function values alone.
Central difference (G-363)	A way to estimate a slope from two nearby points, one small step h on each side: $(f(x + h) - f(x - h))/(2h)$ (a numerical derivative). It is more accurate than a one-sided step.
Product rule (G-1577)	The rule for the derivative of a product of two functions: differentiate one factor at a time, keep the other, and add: $(fg)' = f'g + fg'$.
Quotient rule (G-1609)	The rule for the derivative of a fraction of two functions: $(f/g)' = (f'g - fg')/g^2$.

Term	Meaning
Maclaurin series (G-1142)	The Taylor series around $x_0 = 0$: an infinite sum of powers of x, built from a function's derivatives at 0, that approximates the function near 0.
Linearisation (G-1099)	Replacing a function near a point by its tangent line (its first-order Taylor polynomial), so that a curve can be handled as a straight line close to that point, where straight-line methods apply.
Analytic function (G-197)	A function that can be written exactly as its Taylor series (an infinite sum of powers) near every point.
Power series (G-1541)	A polynomial with infinitely many terms, $\sum a_k(x-c)^k$; the Taylor series is one example, and it lets functions such as e^x and $\sin x$ be computed from polynomials.
Gradient descent (G-862)	Finding the lowest point of a function by repeated small steps downhill.
Chain rule (G-371)	To differentiate a function of a function, multiply the outer derivative by the inner derivative.
Taylor series (G-1954)	Approximation of a function near a point by a polynomial built from its derivatives there.
Composition (G-431)	Applying one transformation and then another, seen as one overall transformation.
Error (residual) (G-705)	The gap between an actual value and the model's prediction on one data point: actual minus predicted.
Second derivative (G-2249)	The derivative of the derivative; it measures how the curve bends.